

COMPUTER LOGIC AND ORGANIZATION

CDA 3103C

SPRING 2026

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Outline

- ❑ Self-introduction
- ❑ Course Details
- ❑ Course Goals
- ❑ Importance of this Course
- ❑ Course Topics

Background

❑ Education

- ❑ Ph.D., Computer Science at UCF, 2020
- ❑ M.Sc., Computer Science at UCF, 2015
- ❑ B.Sc., Mathematics and Computing at Istanbul Kultur University, 2010

❑ Work

- ❑ Assistant Professor at UCF, 2022-Present
- ❑ Scientific Solutions Engineer at Flywheel.io, 2021-22
- ❑ AI Algorithm Engineer at Roche Molecular Systems, 2020-21
- ❑ Deep Learning Research Scientist Intern at Siemens Healthineers, 2018

❑ Research/Teaching Interests

- ❑ Algorithm Design and Deep Learning for Image Processing
- ❑ Adversarial Attacks and Defenses in Deep Learning
- ❑ Medical Image Analysis

❑ Honors & Awards

- ❑ Best Paper Candidate AISafety at IJCAI, 2019
 - *On the Susceptibility of Deep Neural Networks to Natural Perturbations*
- ❑ Graduated 3rd in department, Istanbul Kultur University, Mathematics and Computer Science, 2010

- ❑ Computational Methods in FinTech I & II
- ❑ Artificial Intelligence for FinTech
- ❑ Web and User Experience
- ❑ Computer Forensics I & II
- ❑ Analysis of Computer Communication Networks
- ❑ Network Science
- ❑ Computer Logic and Organization
- ❑ Network Security and Privacy

Course Details

- **Recommended Course Textbook:**
 - Computer Organization and Design MIPS Edition
 - by David Patterson, 6th Edition
 - Prereqs: COP 3223 (Introduction to Computer Programming)

Course Term: Spring 2026

Course Meetings: Tue & Thu; 12:00 pm - 1:15 pm

Course via Zoom: Live Sessions (Links will be provided.)

Course Modality: V

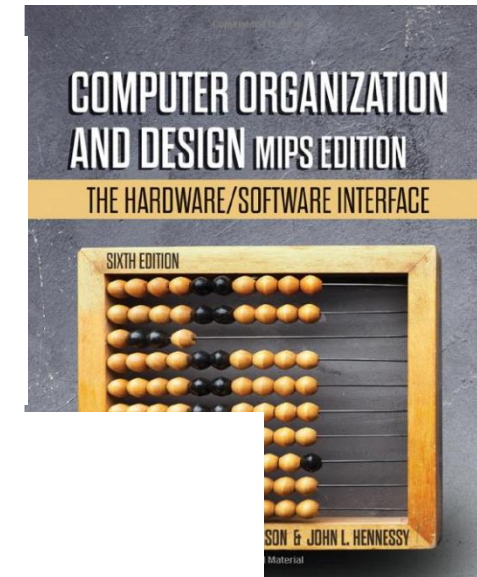
Instructor: Mesut Ozdag, Ph.D.

Office Location: HEC 255

Office Hours: Mo 4:30PM-5:30PM

Office Phone: 407-823-2085

Digital Contact: mesut.ozdag@ucf.edu or message on Webcourses.



Lab (*Recitation*) Sessions:

Fri 12:30PM - 1:20PM TA 0117

Fri 1:30PM - 2:20PM TA 0117

Fri 2:30PM - 3:20PM TA 0117

TBA

GTA:

Office Hours: TBA

Zoom Link: TBA

Lab (*Recitation*) Timeline:

Starts on the second week. Please check announcements.

TBA

TA:

Office Hours: TBA

Zoom Link:

Course Details

- Website: Webcourses, check at least a couple of times each week
- Instructor Contact:
 - Preferred: Webcourse messages (Inbox → Compose Message “Teachers”)
 - Emails
 - Include “CDA3103C” in the subject line.
 - Expect up to 48h (M-F)
 - Resend only after receiving no response after 48h.

Course Details

Assessments and Grading Procedures

<u>Breakdown of Assessments</u>		<u>Grading Scale</u>	
Assessment	Percent of Final Grade	Score out of 100	Letter Grade
Assignments	25%	90 – 100	A
Project	25%	80 – 89	B
Midterm	25%	70 – 79	C
Final Exam	25%	60 – 69	D
Total	100%	0 – 59	F

Course Details: Assignments

- There are about 7 assignments planned.
- *Late assignments* will not be accepted.
- Exceptions apply in case of university-approved absences.
 - “SAS students, illness, serious family emergencies, special curricular requirements (e.g., judging trips, field trips, professional conferences), military obligations, severe weather conditions, and religious holidays.”
 - “participation in official University sponsored activities, such as music performances, athletic competition, or debate”
- Making exception requests as early as possible (preferably 12h before due time) by **emailing the instructor with documents**.
 - Requests made after the due time are ignored automatically.

Course Details: Exams

- Midterm and Final Exams.
- Make-Up Exams
 - University-approved absences
 - Making requests as early as possible (preferably 12h before due time) by **emailing the instructor**.
 - Requests made after the due time are ignored automatically.

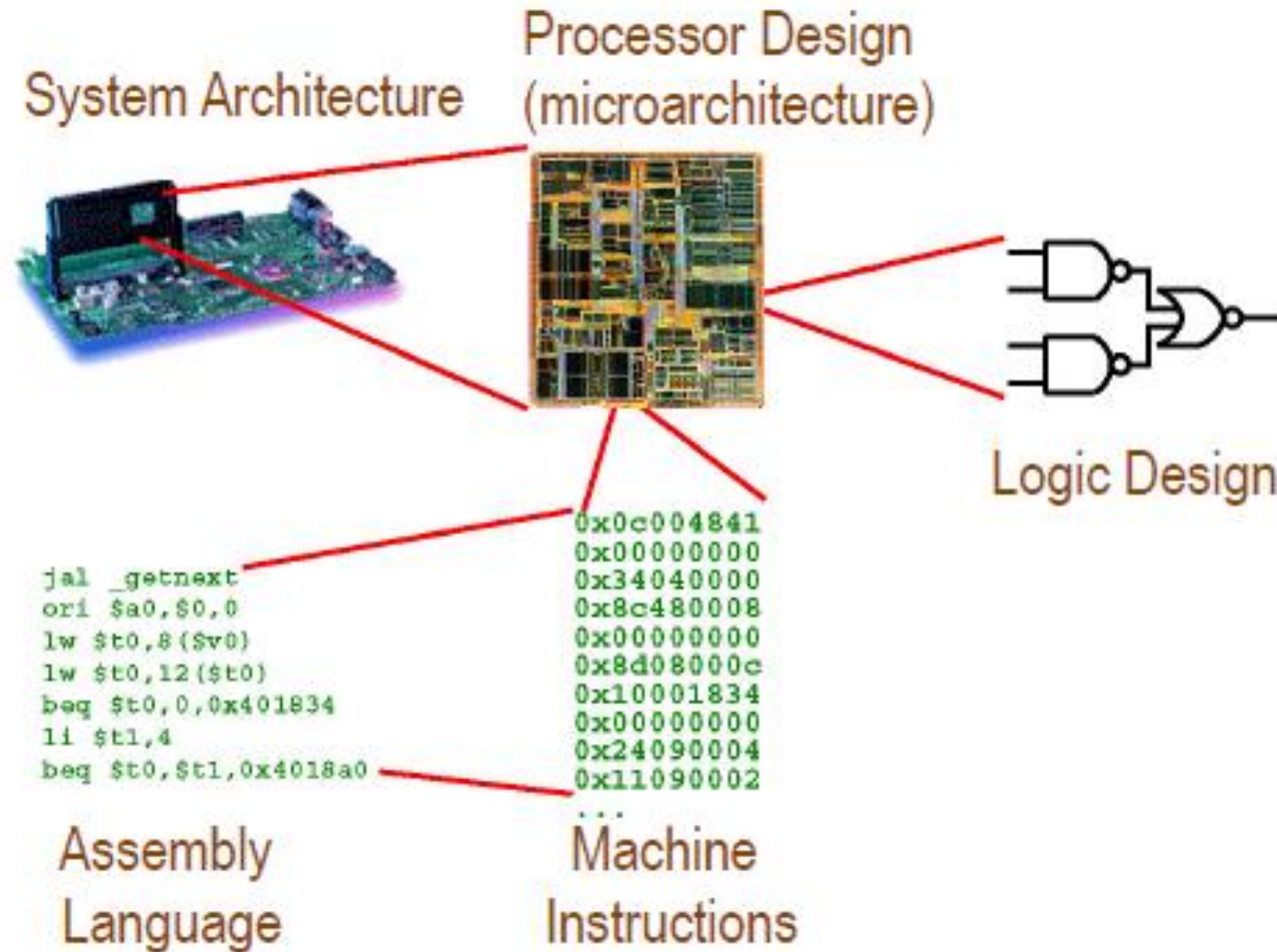
Course Details: Project

- Course project that takes ~3 weeks.
- Topic: TBD

Course Goals

- Upon completing CDA 3103 students will have a deep understanding of the overall organization and design of computers from both programmers' and architects' points of view. Through course assignments and exams, students will calculate binary values, create design specifications for hardware elements, produce assembly language code segments, identify elements of a datapath and their specific interactions, evaluate processor performance and analyze memory behavior.
- Students will be capable of designing, modifying, and simulating a single-cycle data path and its control unit. In a final project, students will produce a functioning simulator of a reduced MIPS processor.
- Students will be familiar with the concepts of pipelining and instruction-level parallelism. Through course assignments and exams, students will analyze segments of assembly code to identify potential hazards and their resolutions which may include forwarding, static branch prediction, dynamic branch prediction, and stalls.
- Students will have an understanding of parallel and distributed computing concepts as they apply to computer processing units.
- Students will have an understanding of the operating system's role in overseeing the computer architecture, interrupt handling, and virtual memory.

CDA 3103 focuses on:



Why should you know these topics?

- These components are important to a number of fields:
 - Operating Systems
 - Compiler design and optimization
 - Mobile and low-power architectures
 - Parallelism
 - Parallel programming
 - Parallel processing
 - Parallel architectures
 - Security
- It's important to know how different parts of the system contribute to the system, interact, and affect each other

Course Topics

- Data Representation
- Boolean Logic Design
- Instruction Set Architecture (ISA)
- MIPS Assembly Language Programming
- Computer Arithmetic (Integer and IEEE 754 Floating Point)
- Performance and Power Metrics
- Datapath Design and Control Logic Design
- Pipelining Basics
- Memory Hierarchy, Caches, and Virtual Memory



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